$$\int \frac{x^3+5x^2-9}{x+2}dx = \int\left[x^2+3x-6+\frac{3}{x+2}\right]dx$$

$$= \frac{x^3}{3}+\frac{3x^2}{2}-6x+3\log|x+2|+c$$

Example 2.13

Evaluate $\int \frac{7x-1}{x^2-5x+6}dx$

Solution:

By partial fractions,

$$\frac{7x-1}{x^2-5x+6}=\frac{A}{x-3}+\frac{B}{x-2}$$

$$\Rightarrow \frac{7x-1}{x^2-5x+6}=\frac{20}{x-3}-\frac{13}{x-2}$$

$$\int \frac{7x-1}{x^2-5x+6}dx = \int\left[\frac{20}{x-3}-\frac{13}{x-2}\right]dx$$

$$= 20\int\frac{dx}{x-3}-13\int\frac{dx}{x-2}$$

$$= 20\log|x-3|-13\log|x-2|+c$$

Example 2.14

Evaluate $\int \frac{3x+2}{(x-2)^2(x-3)}dx$

Solution:

By partial fractions,

$$\frac{3x+2}{(x-2)^2(x-3)}=\frac{A}{(x-2)}+\frac{B}{(x-2)^2}+\frac{C}{(x-3)}$$

$$\Rightarrow \frac{3x+2}{(x-2)^2(x-3)}=-\frac{11}{(x-2)}-\frac{8}{(x-2)^2}+\frac{11}{(x-3)}$$

$$\int \frac{3x+2}{(x-2)^2(x-3)}dx$$

$$= \int\left[-\frac{11}{(x-2)}-\frac{8}{(x-2)^2}+\frac{11}{(x-3)}\right]dx$$

$$= -11\int\frac{dx}{(x-2)}-8\int\frac{dx}{(x-2)^2}+11\int\frac{dx}{(x-3)}$$

$$= -11\log|x-2|+\frac{8}{x-2}+11\log|x-3|+c$$

$$= 11\log\left|\frac{x-3}{x-2}\right|+\frac{8}{x-2}+c$$

Example 2.15

Evaluate $\int \frac{3x^2+6x+1}{(x+3)(x^2+1)}dx$

Solution:

By partial fractions,

$$\frac{3x^2+6x+1}{(x+3)(x^2+1)}=\frac{A}{(x+3)}+\frac{Bx+C}{(x^2+1)}$$

$$\Rightarrow \frac{3x^2+6x+1}{(x+3)(x^2+1)}=\frac{1}{(x+3)}+\frac{2x}{(x^2+1)}$$

$$\int \frac{3x^2+6x+1}{(x+3)(x^2+1)}dx$$

$$= \int\left[\frac{1}{(x+3)}+\frac{2x}{(x^2+1)}\right]dx$$

$$= \int\frac{dx}{(x+3)}+\int\frac{2x}{(x^2+1)}dx$$

$$= \log|x+3|+\log|x^2+1|+c$$

$$= \log\left|(x+3)(x^2+1)\right|+c$$

$$= \log|x^3+3x^2+x+3|+c$$

Exercise 2.2

Integrate the following with respect to x.

1. $\left(\sqrt{2x}-\frac{1}{\sqrt{2x}}\right)^2$
2. $\frac{x^4-x^2+2}{x-1}$
3. $\frac{x^3}{x+2}$
4. $\frac{x^3+3x^2-7x+11}{x+5}$
5. $\frac{3x+2}{(x-2)(x-3)}$
6. $\frac{4x^2+2x+6}{(x+1)^2(x-3)}$
7. $\frac{3x^2-2x+5}{(x-1)(x^2+5)}$
8. If $f'(x)=\frac{1}{x}$ and $f(1)=\frac{\pi}{4}$, then find $f(x)$

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